Ruiyang Liu

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EDUCATION

Nanyang Technological University **2025.8-2026.12 (expected)**
M.Sc. in Computer Control and Automation; CGPA: 4.29 / 5.0 **Singapore**

- School of Electrical and Electronic Engineering
  Supervisor: Prof. Lihua Xie, IEEE Fellow.

Northeastern University (Project 985 University) **2018.9-2022.6**
B.Eng. in Robotic Engineering; Average Score: 82.03 / 100 **Shenyang, China**

- Faculty of Robot Science and Engineering
  Selected Courses: Intelligent Robot Systems Design (95), Robot Perception & HMI (93), Artificial Intelligence (90), C++ (90), Python (85).
- **Honors:** National Champion, 16th National College Student Smart Car Competition (2021.9).
  First Prize, Space Cup Mobile Robot AI Innovation Technology Challenge.
  Outstanding Student Scholarship, Northeastern University (2022.10).

PUBLICATIONS

** Equal contribution.*

[1] Rui Jin*, **Ruiyang Liu***, Xinhang Xu, Yi Wang, Haotian Jin, Yizhuo Yang, and Lihua Xie. QuadHand: A Compact Quadrotor Aerial Manipulator with MRC-SDF-Based Whole-Body Motion Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.*

[2] Xinhang Xu*, **Ruiyang Liu***, Haotian Jin*, Yi Wang, Hongming Shen, Jianping Li, and Lihua Xie. From Sketch Prior to Trajectories: A Mission-Oriented Coordinated Navigation Framework for Indoor UAV Swarm. *International Conference on Control, Automation, Robotics and Vision (ICARCV), 2026.*

[3] Yi Wang*, Rui Jin*, Xinhang Xu*, Haotian Jin, **Ruiyang Liu**, Yizhuo Yang, and Lihua Xie. DuctAM: A Duct-Assisted Quadrotor-Based Aerial Manipulator Enabling High-Force Push-and-Pull Interactions. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.*

[4] Weixiang Guo*, Rui Jin*, Haotian Jin, Xinhang Xu, **Ruiyang Liu**, Haoran Zhao, Yi Wang, Weiqi Gai, Kun Cao, and Lihua Xie. From Local Whole-Body VLA Behaviors to Scene-Scale Aerial Manipulation. *IEEE International Conference on Robotics and Automation (ICRA), 2027. Submitted.*

[5] Jiaye Jin*, Rui Jin*, Xinhang Xu, Haotian Jin, **Ruiyang Liu**, Yi Wang, Jiayan Zhao, Kun Cao, and Lihua Xie. Whole-Body Aerial Grasping and Lifting via a Recurrent Visual Policy. *IEEE International Conference on Robotics and Automation (ICRA), 2027. Submitted.*

RESEARCH EXPERIENCE

Research Focus: Aerial manipulation and whole-body motion planning; multi-UAV navigation; 3D perception and robot learning (RL, recurrent visual policies and VLA).

Ongoing Work: Adaptive planning, grasp-pose prediction and tendon-driven perching.

QuadHand: Aerial Manipulation and Whole-Body Planning | IROS 2026 [1]

- **Platform Development:** Developed a compact quadrotor-based aerial manipulation platform with a 3-DoF arm and gripper, covering mechanical design and hardware integration.
- **Collision-Query Module:** Implemented link-wise MRC-SDF collision queries directly against raw point clouds, preserving fine geometric details for motion planning near obstacles.
- **Whole-Body Planner:** Developed a MINCO-based trajectory planner that jointly optimizes vehicle position, yaw and arm joint motion, incorporating collision costs and dynamic feasibility constraints.
- **Testing & Validation:** Benchmarked the planner in 100 simulated grasping trials, achieving 99% success (SFC: 19%; ESDF: 0%) and 174.22 ms mean optimization time for successful trials. Validated the platform through hardware grasping, cable unplugging and traversal of a 0.6 m × 0.35 m gap.

Sketch-Guided Indoor UAV Coordination | ICARCV 2026 [2]

- **Mapping & Task Representation:** Developed an indoor UAV navigation framework that aligns rough sketch priors with onboard point clouds using ICP and encodes prescribed region sequences.
- **Dynamic Obstacle Trend Layer:** Developed a sliding-window layer to track increases in obstacle occupancy and inflation costs. Used the resulting risk to guide path planning and adapt velocity and acceleration limits.
- **Planning & Coordination:** Integrated weighted A* with MINCO 3D trajectory optimization. Used lightweight region-state exchange and local sensing to coordinate without dense map sharing and continue during communication loss.
- **Validation:** Completed 10/10 simulated missions both with and without communication; validated three-UAV patrols on hardware under 10% packet loss and multi-floor navigation in simulation.

DuctAM: Duct-Assisted Aerial Manipulation | IROS 2026 [3]

- **Control Implementation:** Contributed to implementing and tuning position-attitude control, desired-acceleration decomposition and ducted-fan thrust allocation.
- **Force Testing & Analysis:** Set up and conducted push-pull force tests; analysed airflow-induced measurement interference, trajectory tracking errors and pitch variation to evaluate force-attitude decoupling.

AirGrasp VLA: Scene-Scale Aerial Manipulation | ICRA 2027 (Submitted) [4]

- **Synthetic Data & Policy Execution:** Generated training data from planner trajectories and 3D Gaussian Splatting, pairing synchronized base- and wrist-camera images with action labels. Contributed to measured-progress alignment and continuous trajectory handover using MINCO.
- **Experimental Evaluation:** Conducted latency and command-update-stall tests, scene-graph ablations and hardware experiments; analysed takeover-phase errors, collisions, skill handoffs and task failures.

COMPETITION EXPERIENCE

The 16th National College Student Smart Car Competition (National Top 1) **2020.10-2021.9**

Team Leader **Chongqing, China**

Supported by Chinese Association of Automation

Supervisors: Prof. Zheng Fang (Northeastern University), Assoc. Prof. Shiguang Wen (Northeastern University)

- **Team Leadership:** Led the Aerospace Smart Logistics team to the 2021 national championship, scoring 175 points among 52 finalist teams. Integrated localization, planning, perception and task scheduling to execute dispatch commands for loading, cargo transport and unloading while handling traffic lights, speed bumps and dynamic obstacles.
- **System Integration & SLAM:** Built the vehicle hardware-software architecture, including kinematic analysis, embedded motor control and sensor data acquisition. Benchmarked GMapping, Karto and Cartographer for real-time SLAM, then integrated Cartographer with submap optimization and loop closure for robust localization on the competition track.
- **Planning & Task Execution:** Developed a two-level navigation system using A* for global routes and L1 guidance with DWA for local tracking and obstacle avoidance under kinematic constraints. Designed a finite-state machine for mission-stage transitions and applied low-pass filtering to stabilize paths through S-shaped roads and speed bumps.
- **Perception & Deployment:** Developed HSV-based traffic-light detection and completed camera calibration, data collection and labeling for lane perception. Compared YOLOv5 with lightweight U-Net, tuned the deployed U-Net for varied lighting and used TensorRT for onboard inference, achieving approximately 90% lane detection accuracy in real-track validation.

WORK EXPERIENCE

KEENON Robotics **2022.7-2025.7**

Navigation Algorithm Engineer **Shanghai, China**

- **Edge-Following Planning:** Led development of edge-following functions for commercial cleaning robots operating near boundaries and dynamic obstacles. Combined reference-line planning with a hierarchical state machine to adapt planning and tracking to local scene conditions. Achieved tracking error below 2 cm, reduced collisions by 80%, increased cleaning coverage by 10%, and improved both peak and average speeds by over 30%.
- **Route Teaching & Trajectory Optimization:** Led development of a demonstration mode for user-defined cleaning routes. Recorded robot poses, assigned route attributes according to obstacle proximity, and applied median filtering to suppress pose jitter. Used Bezier and Dubins curves to generate smooth, executable paths. Completed development within one month, supporting over 500 commercial orders.
- **Narrow-Passage & Omnidirectional Navigation:** Optimized restaurant navigation with narrow gaps and overhead obstacles; enabled 5 cm clearance on one side with collision/scraping rate <0.003% for SoftBank deployment. Led omnidirectional obstacle avoidance for a dual-steering-wheel chassis; extended DWA and achieved 40% faster straight-corridor navigation than differential-drive methods.
- **Cleaning Coverage:** Introduced local path offsets and multi-goal selection to reduce delays around obstacles and handle unreachable zone-entry points. Designed a terminal rotation controller to reduce missed boundary areas during zigzag cleaning. Achieved 99% field-task success, peak cleaning efficiency of 1,100 m²/h and 95% effective coverage.

SKILLS & CERTIFICATES

Languages: English (Proficient, IELTS 6.5), Chinese (Native)
Programming: Proficient in C++; familiar with Python. C++ coding standards; Google Test (gtest), unit and white-box testing. Skilled in using Claude and Codex for AI-assisted programming, with code review and test-based validation.
Robotics & Tools: ROS communication and navigation; Stage, Gazebo, Isaac Lab and MuJoCo simulation; OpenCV/PCL for perception; Eigen/g2o for optimization; TensorRT deployment; 3D Gaussian Splatting for rendering and synthetic data.
Development: Linux-based development, deployment and debugging on robot platforms; Git and Gerrit for version control, branch management and collaborative code review.